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MACHINE LEARNING FOR EARTH OBSERVATION

CONCEPT NOTE

Estimating Urban Property Values through Neighbourhood Physical Quality: A Machine Learning and Sentinel-2 Approach in Kicukiro District

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Introduction

Rapid urbanization is reshaping cities across sub-Saharan Africa at a quick pace, and Kigali stands among the most striking examples of this transformation. Rwanda's capital has experienced sustained population growth and infrastructure expansion over the past two decades, drawing both domestic rural migrants and international investment into its urban economy. According to the Rwanda National Institute of Statistics (National Institute of Statistics of Rwanda, 2022), Kigali's population is projected to reach 3.8 million by 2030, placing intense pressure on land markets across all three of the city's administrative districts. Within this environment of growth, land prices in Kigali have risen sharply, with prime areas recording price increases of more than 300% between 2010 and 2022(World Bank, 2023).

Kicukiro District, located in the south-eastern part of Kigali City and covering approximately 167 km² across 10 administrative sectors, presents a particularly compelling case study of this urban transformation. The district contains a striking contrast of built environments within a compact geographic area: planned, well-serviced residential and commercial zones in sectors such as Kagarama and Niboye in the west, transitional mixed-use areas in Kanombe and Kicukiro Centre, and rapidly densifying informal settlements in eastern sectors such as Gatenga and Busanza (City of Kigali, 2020). This makes Kicukiro an ideal laboratory for studying the relationship between the physical characteristics of urban space and the land values those characteristics generate.

While property prices are shaped by many economic and social factors, the physical quality of the built environment, measured through indicators such as vegetation cover, building density, and spatial order, remains one of the most consistent and spatially explicit drivers of land value (Bockarjova et al., 2020). Satellite remote sensing offers a scalable method to measure precisely these physical indicators. Sentinel-2 multispectral imagery, with its 10-metre spatial resolution and freely available archive, can capture vegetation density through the Normalized Difference Vegetation Index (NDVI), built-up intensity through the Normalized Difference Built-up Index (NDBI), and the spatial texture of urban surfaces through Grey-Level Co-occurrence Matrix (GLCM) metrics(Aryal et al., 2023; Xu et al., 2022). When fused and classified using a machine learning algorithm such as Random Forest, these features can produce a spatially explicit map of neighbourhood physical quality that serves as a practical proxy for property value zonation.(Mutale et al., 2024)

This project applies this approach to Kicukiro District, producing what would be the first machine learning-based Neighbourhood Physical Quality Map for the district, a tool with direct relevance for property valuation, infrastructure investment planning, and equitable local revenue collection.

Problem Statement

Despite Kigali's rapid urban growth and the well-documented surge in land values across its districts, property valuation in Rwanda continues to rely on periodic manual ground surveys carried out by valuation officers from the City of Kigali and the Rwanda Revenue Authority (RRA). These surveys are time-consuming, geographically incomplete, and conducted infrequently, creating a persistent gap between officially recorded property values and the on-the-ground reality. In a city growing as fast as Kigali, this gap widens each year and produces two critical governance failures.

The first failure is fiscal inequity. When sectors that have physically upgraded, through new road surfacing, densification, or private construction investment, remain assessed at outdated values, the city loses legitimate tax revenue, and the tax burden falls unevenly across residents and landowners. Sectors in Kicukiro such as Niboye, which have seen significant formal housing development in recent years, may be underrepresented in the current valuation registry relative to their actual physical condition (City of Kigali, 2020).

The second failure is planning inefficiency. Without a current, city-wide spatial picture of physical neighbourhood quality, planners and infrastructure agencies cannot systematically identify which areas are most in need of road rehabilitation, drainage investment, green space provision, or basic service upgrading. Decisions that should be spatially targeted end up being made on an ad hoc or politically driven basis.

There is currently no automated, satellite-based tool that gives Kicukiro District planners a rapid, repeatable, and low-cost classification of neighbourhood physical quality. Although Sentinel-2 imagery provides sufficient spatial detail to distinguish the physical characteristics that separate a planned residential neighbourhood from a dense informal settlement, this data source has not yet been applied to property value proxy mapping in Kicukiro. This project fills that gap directly by developing a Random Forest classifier trained on Sentinel-2 spectral indices and GLCM texture features, producing a district-wide Neighbourhood Physical Quality Map that can be updated seasonally and used immediately by city planning and revenue authorities.

Research Questions and Objectives

This project is guided by three core research questions:

- How is neighbourhood physical quality spatially distributed across the 10 sectors of Kicukiro District, and do these Levels correspond to known patterns of land value variation within the district?
- How accurately can a Random Forest classifier train on Sentinel-2 spectral indices and GLCM texture features map neighbourhood physical quality Levels across Kicukiro District?
- What are the planning and policy implications of the resulting Physical Quality Map for property valuation prioritization, infrastructure investment targeting, and local revenue collection in Kicukiro?

To answer these questions, the following specific objectives will be pursued:

- To preprocess Sentinel-2 Surface Reflectance imagery for Kicukiro District in Google Earth Engine using cloud masking via the QA60 band, dry-season compositing (June–September 2023), reflectance scaling, and boundary clipping to the district's 10-sector extent.
- To compute two spectral indices NDVI and NDBI and two GLCM texture features, entropy and variance, and evaluate their combined contribution to distinguishing neighbourhood physical quality Levels across the study area.
- To train and evaluate a Random Forest classifier using 70% training, 15% validation, and 15% test data split, and to report model performance using Overall Accuracy, Cohen's Kappa coefficient, and F1-score per quality Level.
- To produce a full-district Physical Quality Map at 10-metre resolution and interpret its results in terms of property valuation support, infrastructure deficit identification, and planning prioritization for Kicukiro District.

Data to be Used and Methodology

To ensure that the entire workflow is reproducible, scalable to other Kigali districts, and executable at zero data cost, this project will rely exclusively on publicly available satellite data processed through the **Google Earth Engine (GEE)** cloud computing platform.

Table 1: Data Sources and Tools to be Used in this Project

Data Source	Dataset / Tool	Resolution	Role in Study
Sentinel-2 MSI Level 2A	COPERNICUS/S2_SR_HARMONIZED (Bands B3, B4, B8, B11, B12)	10 m	Primary satellite imagery for all spectral indices and RF classification phases
Kicukiro Admin Boundary	Rwanda NISR Sector/Cell Shapefiles Kicukiro District (OpenAfrica)	Vector	Official Kicukiro District boundary for image clipping and sector-level aggregation
Field Reference Data	Manually digitised training points in GEE across known Kicukiro neighbourhoods	Point-level	Ground truth labels for model training and accuracy validation across 4 quality Levels
Google Earth Engine	Cloud computing platform (Python API via geemap + ee library)	Cloud-based	Image preprocessing, spectral index computation, GLCM extraction, classification and export

Sentinel-2 was selected as the primary data source because it provides 10-metre resolution in key visible and near-infrared bands, free global coverage with a revisit cycle, and a harmonized Surface Reflectance product that ensures consistency across scenes processed under different ESA baseline versions(Phiri et al., 2020). Imagery will be filtered to the dry season (June–September 2023) to minimize cloud contamination over the Kigali highlands, where cloud cover drops significantly during this period.

The methodology follows five sequential phases, as illustrated in Figure 1 below. Each phase builds directly on the output of the previous one, forming a reproducible pipeline from raw imagery to planning-ready map output.

Figure 1: Five-Phase Methodology Workflow for Neighbourhood Physical Quality Mapping in Kicukiro District

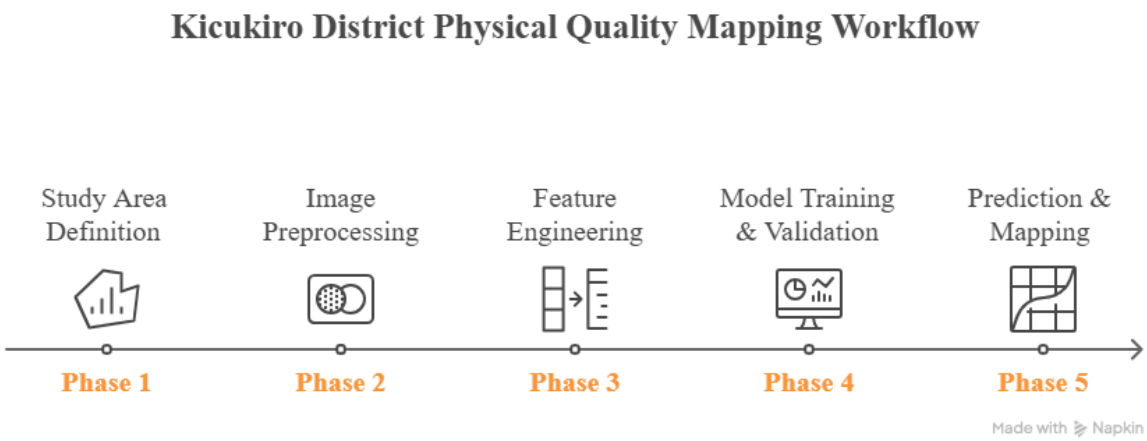

Figure 1: Phases of the methodology study in the workflow

In *Phase 2*, clouds will be masked using Sentinel-2's QA60 quality band, which flags opaque cloud cover at bit 10 and thin circus at bit 11. Reflectance values will be scaled from raw integer format to the 0–1 range by dividing by 10,000. A median composite will then be generated from all qualifying dry-season scenes to produce a single spatially continuous image.

In *Phase 3*, NDVI will be computed as (B8−B4)/(B8+B4) to capture vegetation density and greenness; values above 0.5 are characteristic of well-vegetated, planned residential areas(Altıner & Bingöl, 2025).

NDBI will be computed as (B11−B8) / (B11+B8) to identify built-up surface intensity, with high values indicating dense rooftop and paved surface coverage (Song et al., 2021).GLCM texture features such entropy and variance, will be calculated on Band 8 at a 3×3-pixel window. Entropy captures the degree of spatial randomness in the surface pattern: high entropy reflects the chaotic, irregular roof arrangements typical of dense informal settlements, while low entropy characterizes the ordered, uniform layouts of planned areas(Muhaimin et al., 2022). Together, these four features will be stacked into a single multi-layer input image for classification.

In Phase 4, a Random Forest classifier will be trained to map neighbourhood physical quality across Kicukiro District. Random Forest is well suited to this task because it handles high-dimensional spectral and texture features robustly through ensemble voting, resists overfitting on small training datasets, and produces feature importance scores that directly reveal which inputs drive the classification (Yan et al., 2024; Zhang et al., 2021). Training reference points will be manually placed in GEE using the True Color composite and NDVI layer as visual guides, covering approximately 300 points across four physical quality levels: Level 1 = High

Quality Planned (Kagarama and Niboye); Level 2 = Medium Quality (Kanombe and Kicukiro Centre); Level 3 = Low-Medium Quality (Gatenga); and Level 4 = Dense Informal (Busanza). Model accuracy will be assessed on the withheld 15% test set using Overall Accuracy, Cohen's Kappa coefficient, and F1-score per level

Expected Results

Table 2: Summary of Expected Outputs, Descriptions, and Performance Targets

Expected Output	Description	Target
Physical Quality Map	Full-district GeoTIFF at 10 m showing 4 quality Levels across all 10 Kicukiro sectors	4 Levels mapped across all 10 sectors of Kicukiro District
Model Accuracy Report	Overall Accuracy, Cohen's Kappa, and F1-score per Level evaluated on the held-out test set	OA > 80%, Kappa > 0.75 (Good to Excellent)
Feature Importance Ranking	RF variable importance scores ranking NDVI, NDBI, and GLCM features by predictive contribution	Top 3 predictors identified and interpreted for planning relevance
Planning Priority Map	Colour-coded sector-level map overlaying Level results with Kicukiro boundaries for planner use	Directly usable by Kigali City valuation and infrastructure planning units

The Random Forest classifier is expected to perform within the "Good" to "Excellent" accuracy range, achieving an Overall Accuracy above 80% and a Kappa coefficient above 0.75. This is consistent with recent literature: studies using Sentinel-2 with Random Forest on GEE have reported Overall Accuracy of 89.1% and Kappa of 0.84 (Svoboda et al., 2022), while vegetation and surface classification combining Sentinel-2 spectral bands with GLCM texture features achieved OA of 90.9% and Kappa of 89% (Mohammadpour et al., 2022).

The Physical Quality Map is expected to reveal a clear west-to-east gradient across Kicukiro District. Level 1 and Level 2 pixels are anticipated in Kagarama and Niboye, where formal housing and green corridors dominate, while Level 3 and Level 4 pixels are expected to concentrate in Gatenga and Busanza, where building density is high, NDVI is consistently low, and GLCM entropy is elevated due to irregular rooftop clustering(City of Kigali, 2020).

From a planning standpoint, the map delivers two direct instruments for district governance: a property valuation priority layer directing Rwanda Revenue Authority survey teams toward fiscally under-assessed sectors, and an infrastructure deficit overlay flagging Level 3 and Level 4 areas for road, drainage, and green space investment. Overall, this project shows that freely available Sentinel-2 imagery and a straightforward ML pipeline can produce a district-level quality map at zero data cost, fully reproducible and scalable to any of Kigali's remaining districts.

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